

Thermoelectric Properties of Solution-Processed n-Doped Ladder-Type Conducting Polymers

Suhao Wang, Hengda Sun, Ujwala Ail, Mikhail Vagin, Per O. Å. Persson, Jens W. Andreasen, Walter Thiel, Magnus Berggren, Xavier Crispin, Daniele Fazzi,* and Simone Fabiano*

Conjugated polymers are an emerging class of materials for large-area solid-state energy conversion and storage applications.^[1] These materials enable new paths toward a more sustainable energy landscape without the need of expensive, or even toxic, metal-based compounds. The possibility to harness heat wasted in our daily life by converting a temperature gradient into electricity via thermoelectric generators (TEGs) is attractive, when compared to the organic Rankine cycle, despite their low efficiency. Indeed, TEG is an electronic device without mechanical parts that wear out and it can be easily scaled down for various applications such as powering sensors.^[2] Recently, conducting polymers have been identified as potential thermoelectric materials for the low temperature range (<200 °C).^[3] Their constituting atomic elements (e.g., C, N, O, S) are from millions to billions of times more abundant than those present in state-of-the-art thermoelectric materials for low temperature applications, for instance BiTeSb alloys. Moreover, polymers are lightweight and allow for mechanically flexible devices. Driven by their versatile chemical synthesis, conducting polymers can be scaled up for mass production of thermoelectric devices at relatively low cost, via room-temperature and solution-based manufacturing processes.^[4] However, building efficient thermoelectric devices requires high-performance complementary p-type (hole-transporting) and n-type (electron-transporting) materials. Up to date, all-organic thermoelectric devices have been difficult to manufacture due to the limitations encountered by the n-type organic semiconductors. Unlike their p-type

counterparts, displaying conductivity up to 1000 S cm⁻¹,^[5] n-doped conducting polymers typically suffer from a low electron conductivity (σ), primarily due to their low electron affinity that strongly restricts the n-doping level. In thermoelectric applications, low σ translates into low power factor (defined as $S^2\sigma$, with S being the Seebeck coefficient) and ultimately into a low thermoelectric figure of merit ($ZT = S^2\sigma/kT$, with k the thermal conductivity and T the temperature). Like for other conductors, heat is transported by phonons and charge carriers in conducting polymers;^[6] yet, their unique feature is an intrinsically low phonon contribution, which leads to low k values (0.3–0.8 W m⁻¹ K⁻¹). The other two parameters defining ZT , i.e., σ and S , are interrelated as a function of the charge carrier concentration of the material, typically featuring opposite trends which lead to a maximum value of $S^2\sigma$ for a certain doping level.^[7] The optimization of σ and S , together with a proper understanding of the charge (i.e., polaron) transport mechanisms, represents the key point in maximizing the thermoelectric properties of conducting polymers. Today, the record values of ZT for p-type conducting polymers are 0.25,^[3] 0.31,^[8] and 0.42^[9] at room temperature.

The best performing n-type organic thermoelectric materials up to date are organometallic polymers, featuring an electrical conductivity as high as 40 S cm⁻¹ and a power factor of up to 66 μ W m⁻¹ K⁻².^[10] However, these materials are not directly processable from solution, thus severely restricting their extensive application. With the exception of just a few polymers based on benzodifurandione-phenylenevinylene derivatives, showing n-type conductivity as high as 14 S cm⁻¹,^[11] solution-processed n-doped conducting polymers show typically conductivities of less than 10⁻² S cm⁻¹. Recently, Chabinyk and co-workers reported a maximum conductivity of $\approx 10^{-3}$ S cm⁻¹ for donor–acceptor polymers, such as poly{[N,N'-bis(2-octyldodecyl)-naphthalene-1,4,5,8-bis(dicarboximide)-2,6-diyl]-alt-5,5'-(2,2'-bithiophene)} [P(NDI2OD-T2)] doped with benzimidazol-based dopants (e.g., 4-(1,3-dimethyl-2,3-dihydro-1H-benzimidazol-2-yl)-N,N-dimethylaniline, N-DMBI).^[12] It is suspected that a poor solubility of the dopant in this polymer matrix results in a low doping level, thus causing the low σ . However, similar σ values have been reported for P(NDI2OD-T2) doped with dimer dopants that react effectively and quantitatively by electron transfer,^[13] which instead indicates an intrinsic upper limit for the electron conductivity of this class of materials. Theoretical and experimental studies on donor–acceptor polymers reveal that, despite the high carrier mobility, polarons are localized on the chains.^[14] This results in a charge transport that mainly occurs along the polymer

Dr. S. Wang, Dr. H. Sun, Dr. U. Ail, Dr. M. Vagin,
Prof. M. Berggren, Prof. X. Crispin, Dr. S. Fabiano
Laboratory of Organic Electronics
Department of Science and Technology
Linköping University
SE-60174 Norrköping, Sweden
E-mail: simone.fabiano@liu.se

Prof. P. O. Å. Persson
Thin Film Physics Division
Department of Physics Chemistry and Biology
Linköping University
SE-581 83, Linköping, Sweden

Dr. J. W. Andreasen
Technical University of Denmark
Department of Energy Conversion and Storage
Frederiksborgvej 399, 4000 Roskilde, Denmark

Prof. W. Thiel, Dr. D. Fazzi
Max-Planck-Institut für Kohlenforschung
Kaiser-Wilhelm-Platz 1, D-45470 Mülheim an der Ruhr, Germany
E-mail: fazzi@mpi-muelheim.mpg.de



DOI: 10.1002/adma.201603731

backbone and is macroscopically supported by short-range intermolecular aggregation.^[15] Recent measurements on novel donor–acceptor perylenediimide- and naphthalenediimide-based polymers suggest a correlation between the delocalization of polarons and the macroscopic conductivity.^[16] However, the relationship between chemical structure, polaron localization/delocalization, and doping efficiency remains unclear.

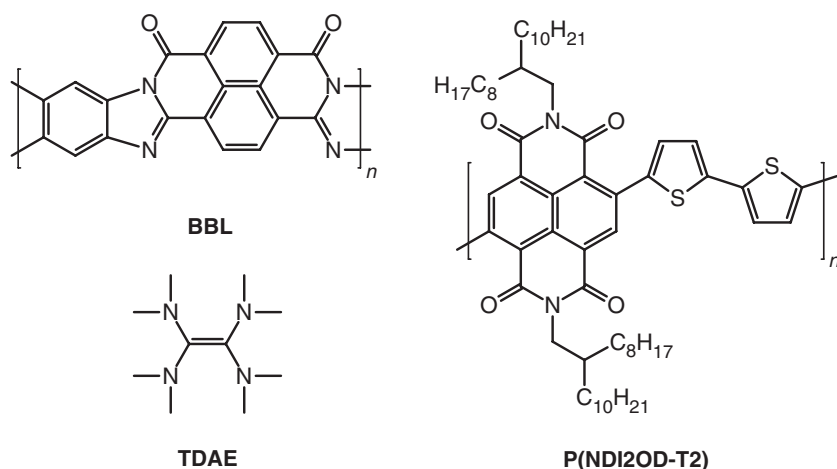
Here we show that ladder-type conducting polymers, such as solution-processable n-type polybenzimidazobenzophenanthroline (BBL),^[17] can achieve n-type conductivities as high as 2.4 S cm^{-1} when doped with strong reducing agents such as tetrakis(dimethylamino)ethylene (TDAE). These remarkable values are three orders of magnitude higher than those measured for P(NDI2OD-T2), here used as the reference model system. Insights into the polaronic properties of both polymers were gained through density functional theory (DFT) calculations. An adequate description of polarons in long BBL oligomers (approaching the polymer limit) requires the use of the unrestricted DFT broken symmetry (UDFT-BS) approach. This aspect, of importance by itself, reflects the stability of polarons in extended π -conjugated systems and suggests that the polaron wavefunction in narrow-bandgap π -conjugated ladder-type polymers is of multiconfigurational character. The computed polaron delocalization length for the linear -“torsion-free”- homopolymer BBL is larger than that of the distorted donor–acceptor polymer P(NDI2OD-T2). According to recent findings by Bao and co-workers,^[16] this might suggest a higher intrachain polaron mobility for BBL than P(NDI2OD-T2). In this frame, the observed high BBL electron conductivity can be already rationalized at the single-chain scale. By carefully modulating the doping levels, we optimized the Seebeck coefficient and power factor, reaching values of $\approx 0.43 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$, which are one order of magnitude higher than those achieved in P(NDI2OD-T2).

Scheme 1 illustrates the chemical structures of BBL, P(NDI2OD-T2), and TDAE. Both BBL and P(NDI2OD-T2) have several compelling properties, including good solution processability, air stability, and high field-effect mobilities.^[17,18] However, unlike P(NDI2OD-T2), BBL is not a donor–acceptor polymer and, most notably, it shows a highly rigid and planar

polymer backbone, which leads to unusual high glass transition temperature ($>500 \text{ }^\circ\text{C}$) and high thermal stability.^[17] TDAE was chosen as the n-dopant, since it has successfully been used for the optimization of the thermoelectric figure of merit of p-type conducting polymers such as poly(3,4-ethylenedioxythiophene).^[13,19] It is then appealing to explore this material in the fabrication of printed thermoelectric generators, as it allows for the simultaneous optimization of oxidation levels in both p- and n-doped polymers.

To analyze the changes associated with the doping level of the n-type polymers exposed to TDAE vapors, UV–vis–NIR spectroscopy was performed on thin films for different exposure times. Figure 1a shows optical absorption spectra of BBL films before and after TDAE exposure. An intense absorption peak (A) at around 580 nm is clearly evident for the pristine film, similarly to what has previously been reported for pristine (undoped) BBL films.^[20] Based on our TD(DFT) calculations performed on BBL oligomers featuring different repeat units (BBL_n , $n = 1, 2, 4, 8$, see the Supporting Information), band A is associated with the S_0 – S_1 transition, with S_1 being a Frenkel-like exciton delocalized over up to six repeat units. After TDAE treatment, a clear reduction of band A is observed together with the appearance of a new broad band (B) at about 900 nm. TD(UDFT-BS) calculations performed on the longer oligomer (BBL_8 , vide Figure 1c and Supporting Information) well reproduce the intensity reduction and broadness of band A, and the appearance of the new band B, ascribed to polaron-induced transitions. Although DFT calculations on oligomer models tend to overestimate the transitions energies,^[21] the computed spectral shape and band intensity ratio are consistent with the experimental trends; furthermore, our assignments are also corroborated by previous reports on electrochemically doped BBL thin films,^[20] thus confirming the successful doping of the polymer.

Figure 1b reports the UV–vis–NIR spectra of pristine and TDAE-doped P(NDI2OD-T2) thin films. In the undoped pristine state, P(NDI2OD-T2) shows the typical absorption features already reported in literature,^[22] which are assigned to a high-energy π – π^* transition (D') at 390 nm and a broad, low-energy band (A') centered at 705 nm, mainly associated with an intramolecular charge transfer between the naphthalene dicarboximide moiety and the bithiophene units. This structured low-energy absorption feature also represents a clear spectroscopic fingerprint of aggregated species. N-doping is accompanied by a quenching of band A' , along with the rising of two absorption bands (C' and B') at roughly 500 and 820 nm, respectively. In addition, band D' shifts toward longer wavelengths (Figure 1b) as the n-doping level increases. From TD(UDFT) calculations conducted on long charged P(NDI2OD-T2) oligomers ($n = 4, 5$), we can assign the absorption bands of doped films to polaronic species (see Figure 1d and details in the Supporting Information). Our measured spectra and computational assignments are also consistent with the formation of radical anion



Scheme 1. Chemical structure of BBL, P(NDI2OD-T2), and TDAE.

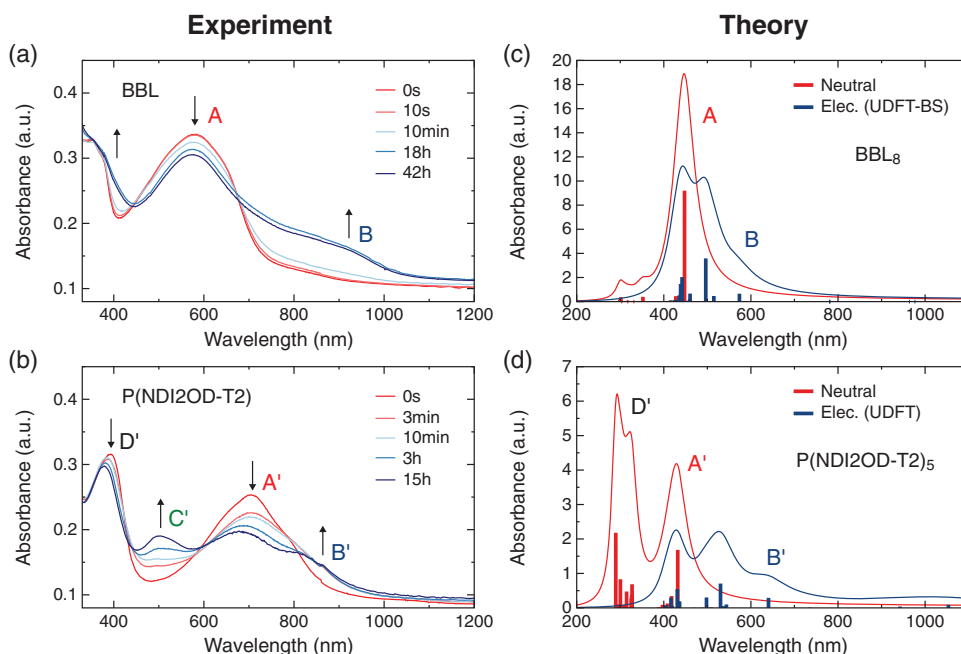


Figure 1. Experimental (left) and calculated (right) UV-vis-NIR absorption spectra of a,c) BBL and b,d) P(NDI2OD-T2) films for different TDAE exposure times (panels a,b) and for neutral and charged electronic states (panels c,d).

species, as found by in situ spectroelectrochemistry^[23] and charge modulation spectroscopy measurements,^[24] assuring the successful doping of P(NDI2OD-T2).

The electrical properties of BBL films were investigated before and after exposure to TDAE vapors in an inert environment. **Figure 2a** shows the electrical conductivity (σ) of n-doped BBL film as a function of the exposure time (i.e., n-doping time). Before exposure, BBL is in its undoped pristine state and shows an electrical conductivity as low as $\approx 1 \times 10^{-7} \text{ S cm}^{-1}$. After exposure to TDAE vapors (3 h, at room temperature), the electrical conductivity dramatically increases, reaching a value as high as $1.7 \pm 0.6 \text{ S cm}^{-1}$, almost seven orders of magnitude higher than that of the pristine state. These values approach those reported for electrochemically doped BBL films (about 2 S cm^{-1}),^[20]

indicating a strong chemical doping ability of TDAE. Interestingly, no changes in the final electrical conductivity are observed for longer exposure times ($>20 \text{ h}$) and even after several weeks of storage in nitrogen. The large increase in the electrical conductivity is in agreement with the rise of polaron absorption bands in the UV-vis-NIR spectra, corroborating the successful n-doping of BBL films. Compared to chemically n-doped P(NDI2OD-T2), BBL shows an electrical conductivity up to three orders of magnitude larger (see Figure 2a). Upon n-doping, P(NDI2OD-T2) conductivity reaches a maximum of $5 \times 10^{-3} \text{ S cm}^{-1}$, which then monotonically decreases for longer exposure time. The observed maximum conductivity value and trend for P(NDI2OD-T2) are comparable to previous reports,^[12,13,16] assuring that films here investigated are

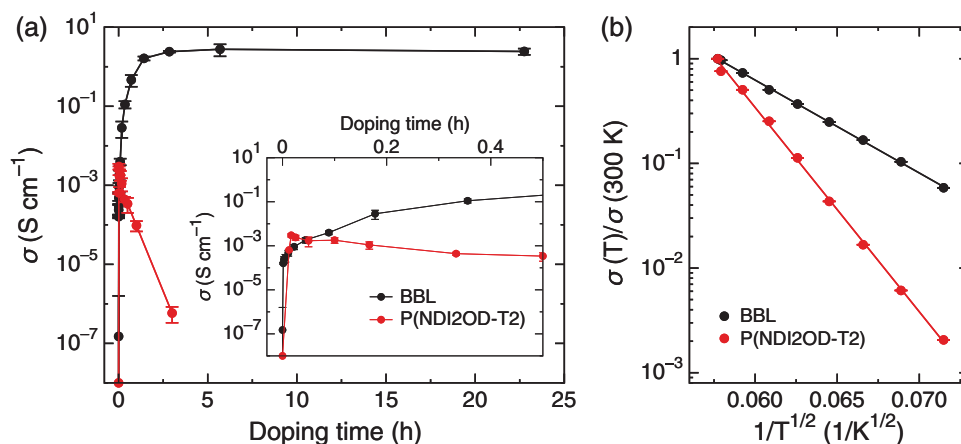


Figure 2. a) Electrical conductivity of BBL and P(NDI2OD-T2) films as a function of the TDAE exposure time (doping time). b) Temperature dependence of the electrical conductivity (normalized to 300 K) for doped BBL and P(NDI2OD-T2) samples.

representative of a high-performance material. Note that the oxidation potential of TDAE (≈ -1 V vs Fc/Fc^+)^[25] is comparable to the reduction potential of the BBL and P(NDI2OD-T2) (see Figure S1, Supporting Information). Thus, the doping more likely takes place through electron transfer from the dopant to the semiconducting polymer. As both BBL and P(NDI2OD-T2) show equal electron affinities ($E_A = -4.0$ eV),^[26] we exclude that the difference in the electrical conductivity of the two polymers, upon doping, is due to different energy-level values.

The maximum electrical conductivity of doped BBL and P(NDI2OD-T2), normalized to 300 K, is plotted in Figure 2b as function of the inverse temperature from 200 to 300 K. The temperature dependence of conductivities is in agreement with a quasi-1D hopping transport (see Figure S2, Supporting Information). At high temperatures, the transport is through nearest neighbor hopping and the corresponding Arrhenius activation energies (E_A) are found to be 0.28 and 0.12 eV for P(NDI2OD-T2) and BBL, respectively. The former is comparable to prior results of P(NDI2OD-T2) doped with DMBI^[12] or $[\text{RhCp}_2]_2$,^[27] and is in agreement with theoretically predicted values derived in the polymer limit.^[14]

The difference in the electrical conductivity between BBL and P(NDI2OD-T2) can be ascribed to their intrinsic (i.e., molecular) structural and polaronic properties. We compared the optimized UDFT structures for the longest charged oligomers (i.e., anions) considered in this study, namely BBL_8 and P(NDI2OD-T2)_5 (Supporting Information). In the charged state, BBL has a flat-planar structure, maintaining the intramolecular order over a long range. P(NDI2OD-T2) shows, instead, a distorted chain, with pronounced dihedral angles ($\approx 50^\circ$ – 70°) between the single conjugated units. These structural differences may have an effect on the polaron delocalization length and the activation energy for charge transport, leading to a low E_A for the structurally ordered BBL polaron, whereas high E_A should arise for the more structurally disordered P(NDI2OD-T2).

In the case of BBL oligomers, when the chain length and hence the π -electron delocalization increases (approaching the polymer limit), the unrestricted DFT solution (UDFT) becomes unstable and is thus not capable of describing the polaronic state in a proper manner. For this purpose, the broken symmetry formalism (UDFT-BS) is better suited. More specifically (see details in Supporting Information), while short BBL oligomers ($n = 1, 2$) do not show any DFT instability in the charged state, longer chains ($n = 4, 8$) do. The corresponding stabilization energy ($\Delta E = E(\text{UDFT-BS}) - E(\text{UDFT})$) is -0.27 eV for BBL_4 and -0.42 eV for BBL_8 , respectively. This implies that for long oligomers (here $n = 4$ and 8), representative of the polymer chains, the UDFT-BS approach should be adopted to describe the polaronic properties in the DFT framework. A detailed comparison between UDFT and UDFT-BS results (i.e., electronic transition energies, polaron structural relaxations, etc.) is reported in

the Supporting Information. For the case of P(NDI2OD-T2), instead, no instability in the electronic structure of the charged state is encountered when increasing the oligomer length, confirming that UDFT is an adequate approach for the description of the polaronic properties for such donor–acceptor systems.

In analogy to biradicaloid systems,^[28] the instability occurring at the UDFT level for long BBL oligomers can be traced back to the symmetry-dilemma or symmetry-breaking problems.^[29] It is related to a multiconfigurational character of the ground state wavefunction. In the case of BBL, a fully satisfactory description of the polaron electronic structure would thus require the use of multiconfigurational wavefunction methods or hybrid approaches (e.g., multiconfigurational pair-density functional theory),^[30] which are however still impractical for such large systems with current state-of-the-art computational resources.

In Figure 3, we compare the spin (i.e., α and β) density distributions of the longest BBL ($n = 8$) and P(NDI2OD-T2) ($n = 5$) oligomers here investigated. The polaron is more delocalized in BBL (UDFT-BS) than in P(NDI2OD-T2) (UDFT): the spin density and the structural relaxations (for the latter see the Supporting Information) extend over almost three repeating units in BBL, rather than only one in P(NDI2OD-T2). We believe that this is related to the planar molecular structure of BBL, which allows for more homogeneous structural relaxations induced by the extra electron. In fact, just considering the single repeat polymer unit ($n = 1$) the polaron is fully delocalized in BBL, while in P(NDI2OD-T2) it is mostly localized on the NDI2OD moiety due to the D–A character of the building blocks, thus causing asymmetric relaxations along the chain.^[14,31]

Grazing-incidence wide-angle X-ray scattering (GIWAXS) and transmission electron microscopy (TEM) were performed to investigate the crystalline morphology and the effect of doping on the polymer crystal structures. The TEM image of undoped P(NDI2OD-T2) films reveals a fiber-like structure

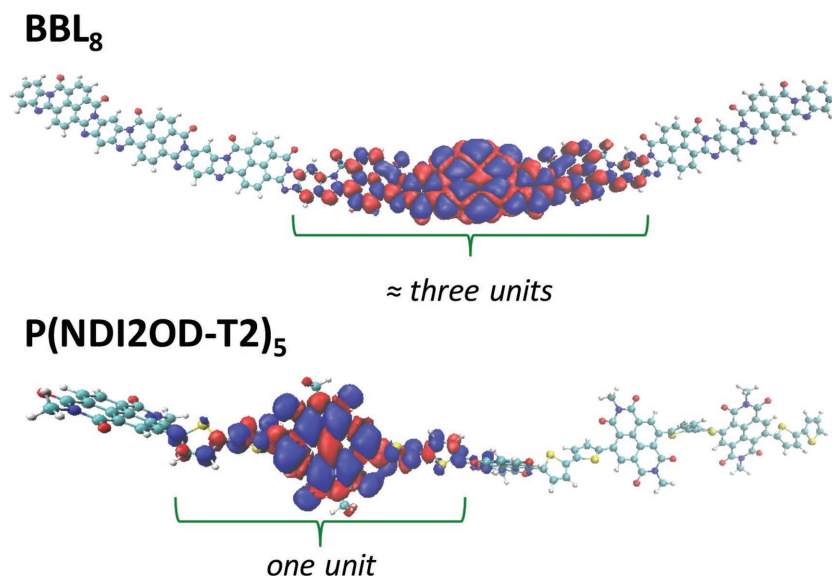


Figure 3. Spin (α and β) density distributions of the longest BBL ($n = 8$, top) and P(NDI2OD-T2) ($n = 5$, bottom) oligomers, as calculated at the UDFT-BS and UDFT level, respectively ($\omega\text{B97X-D3/6-31G}^*$).

qualitatively similar to previously published observations in the neat polymer films.^[32] The GIWAXS is also fully consistent with the known diffraction pattern of P(NDI2OD-T2) and shows a preferential face-on orientation (Figure S3, Supporting Information).^[15] After doping, the fiber-like structure in the TEM image is less visible, while from the GIWAXS the in-plane (100) and (200) lamellar reflections decrease in intensity, such that only the first-order (100) and (001) reflections are visible. In the out-of-plane direction, the (010) peak corresponding to the π - π stacking is smoothed by doping (see Figure S4, Supporting Information). The diffraction pattern of TDAE-doped P(NDI2OD-T2) is qualitatively similar to that recently reported for the same polymer doped with dimer dopants.^[16] In the case of BBL, the GIWAXS data are in good agreement with previously measured diffraction pattern of undoped films.^[33] Upon doping, the texture is not affected, whereas the polymer structure is substantially changed, as evidenced by a rather large shift of the strong (100) lamellar peak, and the concomitant occurrence of an intermediate peak at a lower scattering angle (see Figures S4, Supporting Information). This could be interpreted as a doubling of the unit cell in the direction of the lamellar packing, indicating a change in structure caused by doping. Overall, for both pristine and doped polymers, we observe that BBL exhibits the most pronounced π -stacking, which indicates very high ordering of this polymer.

The Seebeck coefficient of BBL and P(NDI2OD-T2) thin films were also measured as a function of the TDAE exposure time. To minimize the error in determining S , we used an electrode configuration which takes into account the effect of the contact geometry.^[34] The Seebeck coefficient of BBL decreases by a factor of 7 upon exposing the polymer to the TDAE vapors (Figure 4a). S is initially about $-400 \pm 25 \mu\text{V K}^{-1}$ for a σ of around $3 \times 10^{-4} \text{ S cm}^{-1}$, then reaching its minimum of $-60 \pm 4 \mu\text{V K}^{-1}$ for $\sigma = 1 \text{ S cm}^{-1}$. Notably, the sign of the thermovoltage is consistent with electrons being the majority carrier (Figure S5, Supporting Information). In comparison, the Seebeck coefficient of P(NDI2OD-T2) decreases from -790 ± 14 to $-150 \pm 5 \mu\text{V K}^{-1}$ when σ increases from 1×10^{-4} to about $3 \times 10^{-3} \text{ S cm}^{-1}$ (Figure 4a). For long exposure times, σ reduces as observed before, and thus the extraction of S is not straightforward, as the resistance of P(NDI2OD-T2) changes dramatically while heating the substrate (Figure S6, Supporting Information). Note that similar S values are obtained regardless of the film thickness and when *N*-DMBI is used as a dopant agent. Since σ increases while S decreases, at higher doping levels, the power factor ($S^2\sigma$) reaches an optimum at a specific exposure time (doping level). Note that since the electronic contribution to the thermal conductivity is expected to be almost negligible for the range of electrical conductivity in our samples, we only focus our study on the optimization of the power factor. For BBL, a maximum $S^2\sigma$ of $0.43 \mu\text{W m}^{-1} \text{ K}^{-2}$ is obtained, which is one order of magnitude higher than that of P(NDI2OD-T2) ($0.013 \mu\text{W m}^{-1} \text{ K}^{-2}$). Note that the latter is one order of magnitude lower than previously reported for P(NDI2OD-T2), which could be due to contact geometry effect.^[34]

We have attempted to interpret the magnitude of the Seebeck coefficients consistently in terms of variable-range hopping disorder (VRH) model.^[35] Similar analysis has been used to

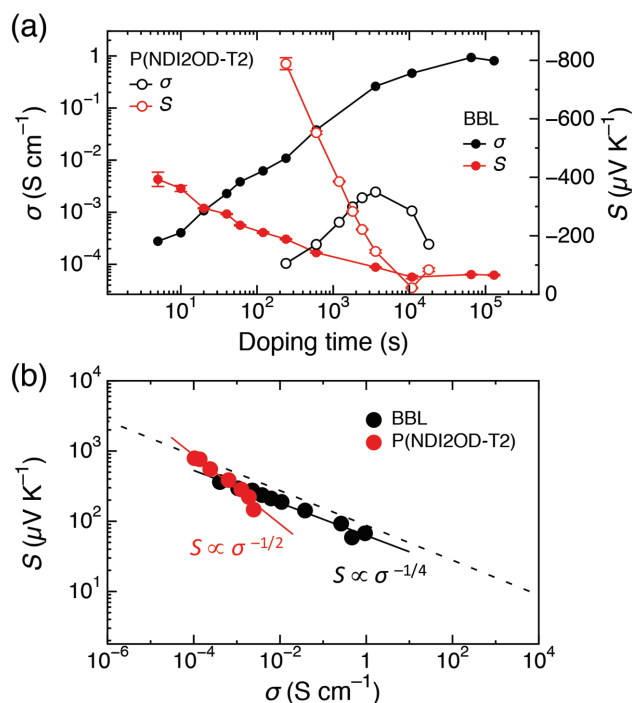


Figure 4. a) Electrical conductivity σ and Seebeck coefficient S versus TDAE doping time for BBL and P(NDI2OD-T2) films. b) Seebeck versus conductivity data for BBL and P(NDI2OD-T2). The dashed line indicates the empirical fit $S \propto \sigma^{-1/4}$ as extracted from ref. [41].

explain comparable measurements in amorphous silicon^[36] and polymers.^[37] The Seebeck coefficient

$$S = \frac{\int (E - E_F) \sigma(E) dE}{eT \int \sigma(E) dE} \quad (1)$$

is determined by the difference between the Fermi energy level (E_F) and the transport energy level (E). In Equation (1), $\sigma(E)$ is the conductivity distribution function, T the temperature, and e the charge of the carrier. In the frame of the VRH model, where the transport is assumed to be dominated by a characteristic hop from the equilibrium energy to a relatively narrow transport energy E^* , Equation (1) becomes

$$S = \frac{(E^* - E_F)}{eT} \quad (2)$$

In the hopping regime, E_F shifts toward the transport energy E^* by increasing the charge carrier density (i.e., the doping level). This explains the typical trend of S of decreasing upon increasing the doping level [i.e., exposure time (see Figure 4a)]. The Seebeck coefficient of P(NDI2OD-T2) is higher than that of BBL, possibly due to a difference in doping level and/or a difference in extension of the charge carrier, as shown in the VRH model.^[38] The potentially lower doping level of P(NDI2OD-T2) compared to BBL is not due to the EA, which is the same for both polymers, but rather to the D-A character of P(NDI2OD-T2), which may cause inefficient doping.^[39]

A second effect rarely discussed but recently pinpointed in a theoretical study^[38] is that the evolution of S with doping level is very dependent on the degree of localization of the carriers. If the localization length is small, S varies quickly with doping level. If the carrier is extended instead, then S varies slowly with doping level.^[38] This means that for equal doping level and equal Gaussian width, the extension of the localized state describing the carrier is affecting the Seebeck coefficient. BBL has a larger polaron extension compared to P(NDI2OD-T2), and this could be one origin of the low S reported in Figure 4a. Note that the weak temperature dependence of S , compared with that of σ (Figure 2b), supports our choice of using a VRH model to interpret qualitatively our results. Considering a VRH (with percolation) model for charge transport, using a Gaussian density of states, the slope of S versus T is related to the degree of disorder.^[40] When BBL and P(NDI2OD-T2) are doped to the maximum conductivity, BBL shows a weaker temperature dependence of S ($\partial S/\partial(1/T) = 0.05$ eV) than P(NDI2OD-T2) (0.13 eV) (see Figure S7, Supporting Information). This may suggest a lower disorder in BBL compared to P(NDI2OD-T2), as indicated by the GIWAXS data.

As a direct measurement of the carrier concentration in organic materials is difficult, due to their low mobility and energetic disorder, it is simplest to discuss the variation of S as a function of σ , as shown in Figure 4b. BBL follows the empirical relation of $S \propto \sigma^{-1/4}$, as already observed for other semiconducting polymers,^[41] while P(NDI2OD-T2) shows a stronger dependence ($S \propto \sigma^{-1/2}$). Although the origin of this empirical trend is not clearly understood yet, the different slope could be due to differences in the molecular order, as observed for stretched polyaniline,^[42] or to a different polaron extension, in agreement with the DFT results.

In conclusion, we demonstrated that linear – “torsion-free” – ladder-type conducting polymers, such as BBL, can reach conductivity values that are three orders of magnitude higher than those of distorted donor–acceptor polymer [e.g., P(NDI2OD-T2)]. This is an important and general material design rule for optimizing the thermoelectric properties of conducting polymers. A realistic DFT description of polarons in long BBL oligomers (approaching the polymer limit) is provided by the broken symmetry approach. The computed polaron and spin delocalization lengths are larger in BBL than in P(NDI2OD-T2), suggesting an easier intramolecular transfer, thus a higher polaron mobility along the chain for the ladder-type polymer. In this frame, the high electron conductivity of BBL can be already rationalized at the single-chain level. The optimized thermoelectric power factor of BBL reaches values that are one order of magnitude higher than those observed for P(NDI2OD-T2). These results provide a simple picture that clarifies the relationship between the backbone structure of the polymer and the polaron delocalization length, setting molecular-design guidelines for the next generation of conjugated polymers.

Experimental Section

Film Preparation and Doping: All devices were fabricated using glass substrates cleaned sequentially in acetone, water, and isopropanol, followed by the drying step with nitrogen. P(NDI2OD-T2) ($M_n = 29.5$ KDa

and PDI = 2.1, Polyera ActivInk N2200, purchased from Polyera Corp.) was dissolved in *o*-dichlorobenzene at the concentration of 5 mg mL⁻¹, and the solution was stirred at 70 °C for ≈ 1 h to allow complete dissolution of the polymer. After stirring, the solution was spin-coated onto the glass substrates at 1000 rpm for 40 s. Prior to doping, the films were thermally annealed at 110 °C under nitrogen atmosphere for 10 min, and cooled down naturally to room temperature. BBL (purchased from Sigma-Aldrich, intrinsic viscosity $[\eta] = 0.58$ dL g⁻¹ in concentrated sulfuric acid at 25 °C^[43]) was dissolved in methanesulfonic acid (MSA) at the concentration of 5 mg mL⁻¹. The solution was stirred at 70 °C for ≈ 1 h to allow complete dissolution of the polymer. After stirring, the solution was spin-coated onto the hexamethyldisilazane (HMDS)-modified glass substrates at 600 rpm for 2 min. The BBL films were then dipped immediately after spin-coating into deionized water to remove MSA. All the BBL films were dried on a hotplate at 70 °C in a glovebox under nitrogen atmosphere overnight after removal from water. Prior to doping, the films were thermally annealed at 110 °C under nitrogen atmosphere for 10 min. The polymer films, deposited on glass substrates, were exposed to the TDAE vapor inside an airtight glass bottle (20 mL in volume filled with 1 mL liquid TDAE), following a procedure reported earlier.^[44] The exposure took place inside a nitrogen-filled glovebox and was halted at the required time by removing the polymeric films from the TDAE bottle.

Optical and Electrical Characterization: Absorption spectra of BBL and P(NDI2OD-T2) films were conducted at room temperature using a UV–vis–NIR spectrophotometer (PerkinElmer Lambda 900). Electrical conductivity and Seebeck coefficient measurements were performed inside a nitrogen-filled glovebox using a semiconductor parameter analyzer (Keithley 4200-SCS). Bottom gate/bottom contacts field-effect mobility was used to extract the electron mobilities of P(NDI2OD-T2) and BBL, that are 1.5×10^{-2} and 2.3×10^{-3} cm² V⁻¹s⁻¹, respectively. For conductivity measurements reported in Figure 2a,b, 12 nm thick Au electrodes with a 3 nm thick Ti adhesion layer ($L/W = 30 \mu\text{m}/1000 \mu\text{m}$) were fabricated on glass substrates prior to active layer deposition. For the Seebeck coefficient measurements reported in Figure 4a for a given doping time at room temperature (thermally evaporated gold electrodes with $L/W = 0.5 \text{ mm}/15 \text{ mm}$), the samples were fixed in between a pair of Peltier modules to maintain the desired temperature difference. Here, ΔT was measured by means of thermocouples. Note that in case of temperature-dependent Seebeck coefficient measurements reported in Figure S7 (Supporting Information) (photolithographically defined electrodes with $L/W = 30 \mu\text{m}/1300 \mu\text{m}$), an integrated heater positioned parallel to the channel electrode/temperature sensors was used to create the temperature difference. The distance between the heater and the hot sensor was 20 μm . In this case, ΔT was measured using gold thermistors. The width of the heater and the thermistors was 20 μm . The Seebeck coefficient was calculated from the shift in the current–voltage curves caused by the thermal voltage ΔT .

Computational Methods: BBL and P(NDI2OD-T2) were modeled through an oligomer approach via DFT calculations using the ω B97X-D3 functional and the 6-31G* basis set. Neutral and charged (i.e., radical-anion) species were described at the restricted and unrestricted DFT level of theory, respectively. The stability of the DFT solution was tested and verified in all cases via the spin-unrestricted broken symmetry (BS) approach.^[45] No instability was found for the neutral species of both polymers and for the charged species of P(NDI2OD-T2), confirming that the restricted DFT treatment is appropriate for these systems. In the case of the charged species of BBL, an instability was found for oligomers with four or more repeat units, for which the UDFT-BS solution was most stable; the stabilization energy is given by the difference $\Delta E = E(\text{UDFT-BS}) - E(\text{UDFT})$. All structures, namely BBL ($n = 1, 2, 4, 8$) and P(NDI2OD-T2) ($n = 4, 5$), were optimized in the neutral and charged states (using their most stable DFT solution). For each oligomer and for each electronic state (neutral and charged), vertical electronic transitions were computed at the TDFT level (or TDUDFT-BS when appropriate). For the description of the polaron species, an assumption of at most one charge per oligomer is made due to the low doping level. All calculations were performed using Gaussian09.^[46]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

S.W., H.S., and U.A. contributed equally to this work. The authors acknowledge support from the Knut and Alice Wallenberg Foundation (project "Tail of the sun") and the Swedish Foundation for Strategic Research (Synergy project). S.F. gratefully acknowledges funding by the Swedish Governmental Agency for Innovation Systems - VINNOVA (Grant No. 2015-04859) and the Advanced Functional Materials Center at Linköping University (Grant No. 2009-00971). D.F. kindly acknowledges the Alexander von Humboldt Foundation for a postdoctoral fellowship. POAP acknowledges the Knut and Alice Wallenberg Foundation for support of the electron microscopy laboratory at Linköping University.

Received: July 14, 2016

Revised: September 20, 2016

Published online:

- [1] a) Y. Liang, Z. Chen, Y. Jing, Y. Rong, A. Facchetti, Y. Yao, *J. Am. Chem. Soc.* **2015**, *137*, 4956; b) A. Facchetti, *Chem. Mater.* **2011**, *23*, 733.
- [2] A. P. Joseph, *IEEE Pervasive Comput.* **2005**, *4*, 18.
- [3] O. Bubnova, Z. U. Khan, A. Malti, S. Braun, M. Fahlman, M. Berggren, X. Crispin, *Nat. Mater.* **2011**, *10*, 429.
- [4] M. Berggren, D. Nilsson, N. D. Robinson, *Nat. Mater.* **2007**, *6*, 3.
- [5] O. Bubnova, Z. U. Khan, H. Wang, S. Braun, D. R. Evans, M. Fabretto, P. Hojati-Talemi, D. Dagnelund, J.-B. Arlin, Y. H. Geerts, S. Desbief, D. W. Breiby, J. W. Andreasen, R. Lazzaroni, W. M. Chen, I. Zozoulenko, M. Fahlman, P. J. Murphy, M. Berggren, X. Crispin, *Nat. Mater.* **2014**, *13*, 190.
- [6] a) J. Liu, X. Wang, D. Li, N. E. Coates, R. A. Segalman, D. G. Cahill, *Macromolecules* **2015**, *48*, 585; b) A. Weathers, Z. U. Khan, R. Brooke, D. Evans, M. T. Pettes, J. W. Andreasen, X. Crispin, L. Shi, *Adv. Mater.* **2015**, *27*, 2101.
- [7] a) A. Shakouri, *Annu. Rev. Mater. Res.* **2011**, *41*, 399; b) G. J. Snyder, E. S. Toberer, *Nat. Mater.* **2008**, *7*, 105.
- [8] S. H. Lee, H. Park, S. Kim, W. Son, I. W. Cheong, J. H. Kim, *J. Mater. Chem. A* **2014**, *2*, 7288.
- [9] G. H. Kim, L. Shao, K. Zhang, K. P. Pipe, *Nat. Mater.* **2013**, *12*, 719.
- [10] Y. Sun, P. Sheng, C. Di, F. Jiao, W. Xu, D. Qiu, D. Zhu, *Adv. Mater.* **2012**, *24*, 932.
- [11] K. Shi, F. Zhang, C.-A. Di, T.-W. Yan, Y. Zou, X. Zhou, D. Zhu, J.-Y. Wang, J. Pei, *J. Am. Chem. Soc.* **2015**, *137*, 6979.
- [12] R. A. Schlitz, F. G. Brunetti, A. M. Glaudell, P. L. Miller, M. A. Brady, C. J. Takacs, C. J. Hawker, M. L. Chabinyc, *Adv. Mater.* **2014**, *26*, 2825.
- [13] B. D. Naab, S. Zhang, K. Vandewal, A. Salleo, S. Barlow, S. R. Marder, Z. Bao, *Adv. Mater.* **2014**, *26*, 4268.
- [14] D. Fazzi, M. Caironi, C. Castiglioni, *J. Am. Chem. Soc.* **2011**, *133*, 19056.
- [15] S. Wang, S. Fabiano, S. Himmelberger, S. Puzinas, X. Crispin, A. Salleo, M. Berggren, *Proc. Natl. Acad. Sci. USA* **2015**, *112*, 10599.
- [16] B. D. Naab, X. Gu, T. Kurosawa, J. W. F. To, A. Salleo, Z. Bao, *Adv. Electron. Mater.* **2016**, *2*, 1600004.
- [17] A. Babel, S. A. Jenekhe, *J. Am. Chem. Soc.* **2003**, *125*, 13656.
- [18] H. Yan, Z. Chen, Y. Zheng, C. Newman, J. R. Quinn, F. Dötz, M. Kastler, A. Facchetti, *Nature* **2009**, *457*, 679.
- [19] H. Wang, J.-H. Hsu, S.-I. Yi, S. L. Kim, K. Choi, G. Yang, C. Yu, *Adv. Mater.* **2015**, *27*, 6855.
- [20] K. Willbourn, R. W. Murray, *Macromolecules* **1988**, *21*, 89.
- [21] a) A. L. D. Laurent, D. Jacquemin, *Int. J. Quantum Chem.* **2013**, *113*, 2019; b) H. Sun, J. Autschbach, *J. Chem. Theory Comput.* **2014**, *10*, 1035.
- [22] M. Schubert, D. Dolfen, J. Frisch, S. Roland, R. Steyrlleuthner, B. Stiller, Z. H. Chen, U. Scherf, N. Koch, A. Facchetti, D. Neher, *Adv. Energy Mater.* **2012**, *2*, 369.
- [23] D. Trefz, A. Ruff, R. Tkachov, M. Wieland, M. Goll, A. Kiri, S. Ludwigs, *J. Phys. Chem. C* **2015**, *119*, 22760.
- [24] M. Caironi, M. Bird, D. Fazzi, Z. H. Chen, R. Di Pietro, C. Newman, A. Facchetti, H. Sirringhaus, *Adv. Funct. Mater.* **2011**, *21*, 3371.
- [25] C. Burkholder, W. R. Dolbier, M. Médebielle, *J. Org. Chem.* **1998**, *63*, 5385.
- [26] a) M. M. Alam, S. A. Jenekhe, *Chem. Mater.* **2004**, *16*, 4647; b) S. Fabiano, H. Yoshida, Z. Chen, A. Facchetti, M. A. Loi, *ACS Appl. Mater. Interfaces* **2013**, *5*, 4417.
- [27] Y. B. Qi, S. K. Mohapatra, S. B. Kim, S. Barlow, S. R. Marder, A. Kahn, *Appl. Phys. Lett.* **2012**, *100*, 4.
- [28] a) D. Fazzi, E. V. Canesi, F. Negri, C. Bertarelli, C. Castiglioni, *ChemPhysChem* **2010**, *11*, 3685; b) J. Casado, R. Ponce Ortiz, J. T. Lopez Navarrete, *Chem. Soc. Rev.* **2012**, *41*, 5672; c) M. Nakano, B. Champagne, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **2016**, *6*, 198.
- [29] a) J. P. Perdew, A. Savin, K. Burke, *Phys. Rev. A* **1995**, *51*, 4531; b) C. D. Sherrill, M. S. Lee, M. Head-Gordon, *Chem. Phys. Lett.* **1999**, *302*, 425; c) P.-O. Löwdin, *Phys. Rev.* **1955**, *97*, 1474.
- [30] G. Li Manni, R. K. Carlson, S. Luo, D. Ma, J. Olsen, D. G. Truhlar, L. Gagliardi, *J. Chem. Theory Comput.* **2014**, *10*, 3669.
- [31] D. Fazzi, M. Caironi, *Phys. Chem. Chem. Phys.* **2015**, *17*, 8573.
- [32] C. J. Takacs, N. D. Treat, S. Krämer, Z. Chen, A. Facchetti, M. L. Chabinyc, A. J. Heeger, *Nano Lett.* **2013**, *13*, 2522.
- [33] A. L. Brisen, S. C. B. Mannsfeld, P. J. Shamberger, F. S. Ohuchi, Z. Bao, S. A. Jenekhe, Y. Xia, *Chem. Mater.* **2008**, *20*, 4712.
- [34] S. V. Reenen, M. Kemerink, *Org. Electron.* **2014**, *15*, 2250.
- [35] W. C. Germs, K. Guo, R. A. J. Janssen, M. Kemerink, *Phys. Rev. Lett.* **2012**, *109*, 016601.
- [36] H. Overhof, W. Beyer, *Philos. Mag. Part B* **1981**, *43*, 433.
- [37] a) Y. Xuan, X. Liu, S. Desbief, P. Leclère, M. Fahlman, R. Lazzaroni, M. Berggren, J. Cornil, D. Emin, X. Crispin, *Phys. Rev. B* **2010**, *82*, 115454; b) D. Venkateshvaran, M. Nikolka, A. Sadhanala, V. Lemaire, M. Zelazny, M. Kepa, M. Hurhangee, A. J. Kronemeijer, V. Pecunia, I. Nasrallah, I. Romanov, K. Broch, I. McCulloch, D. Emin, Y. Olivier, J. Cornil, D. Beljonne, H. Sirringhaus, *Nature* **2014**, *515*, 384.
- [38] S. Ihnatsenka, X. Crispin, I. V. Zozoulenko, *Phys. Rev. B* **2015**, *92*, 035201.
- [39] D. Di Nuzzo, C. Fontanesi, R. Jones, S. Allard, I. Dumsch, U. Scherf, E. von Hauff, S. Schumacher, E. Da Como, *Nat. Commun.* **2015**, *6*, 6460.
- [40] S. D. Baranovskii, I. P. Zvyagin, H. Cordes, S. Yamasaki, P. Thomas, *Phys. Status Solidi (B)* **2002**, *230*, 281.
- [41] A. M. Glaudell, J. E. Cochran, S. N. Patel, M. L. Chabinyc, *Adv. Energy Mater.* **2015**, *5*, 1401072.
- [42] N. Mateeva, H. Niculescu, J. Schlenoff, L. R. Testardi, *J. Appl. Phys.* **1998**, *83*, 3111.
- [43] P. Bornoz, M. S. Prévot, X. Yu, N. Guijarro, K. Sivula, *J. Am. Chem. Soc.* **2015**, *137*, 15338.
- [44] S. Fabiano, S. Braun, X. Liu, E. Weverberghs, P. Gerbaux, M. Fahlman, M. Berggren, X. Crispin, *Adv. Mater.* **2014**, *26*, 6000.

- [45] F. Neese, *J. Phys. Chem. Solids* **2004**, *65*, 781.
- [46] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery Jr., J. E. Peralta, F. É. B. Ogliaro, M. J. Bearpark, J. Heyd, E. N. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. P. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, N. J. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, É. Á. É. N. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, D. J. Fox, *Gaussian 09, Revision D.01*, Gaussian, Inc., Wallingford, CT, USA **2009**.